



deeplearning.ai

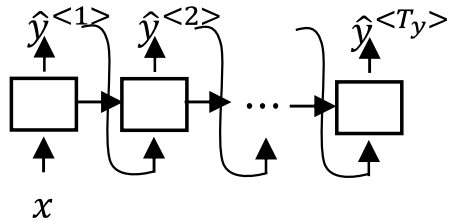
Sequence to
sequence models

Transformers
Intuition

Transformers Motivation

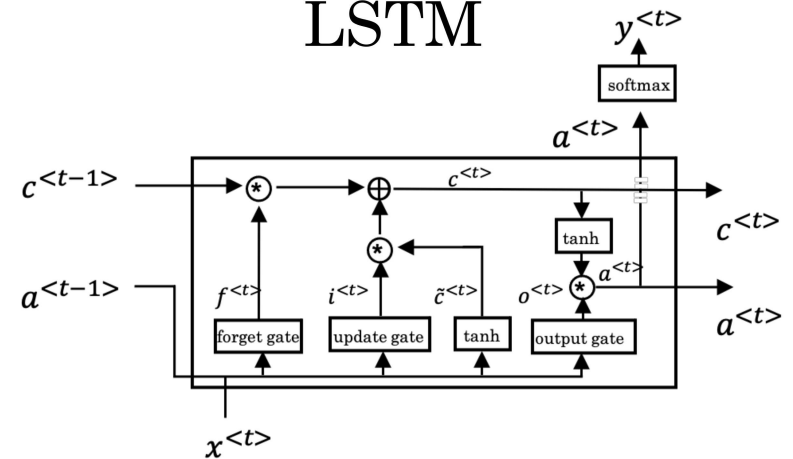
Increased complexity,
sequential

RNN



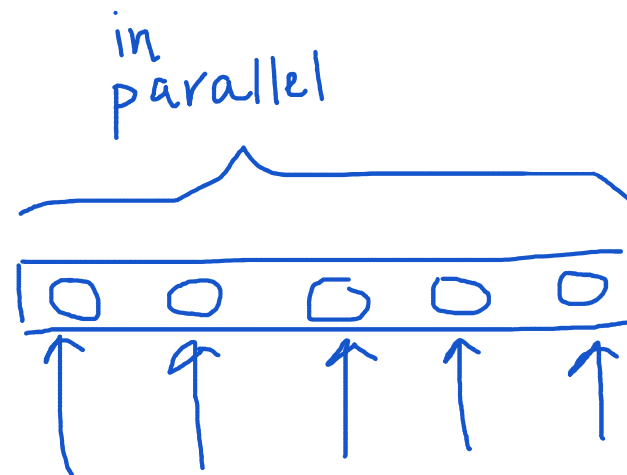
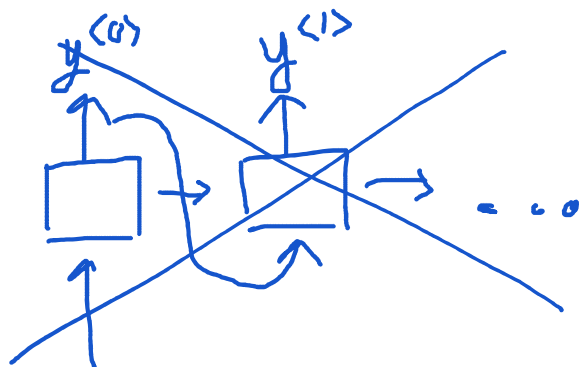
GRU

LSTM



Transformers Intuition

- Attention + CNN
 - Self-Attention
 - Multi-Head Attention





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Sequence to sequence models

Self-Attention

Self-Attention Intuition

$A(q, K, V)$ = attention-based vector representation of a word
→ calculate for each word

RNN Attention

$$\alpha^{<\cancel{t}, t'>} = \frac{\exp(e^{<t, t'>})}{\sum_{t'=1}^T \exp(e^{<t, t'>})}$$

$x^{<1>}$
Jane

$x^{<2>}$
visite

$x^{<3>}$
l'Afrique

$x^{<4>}$
en

$x^{<5>}$
septembre

Transformers Attention

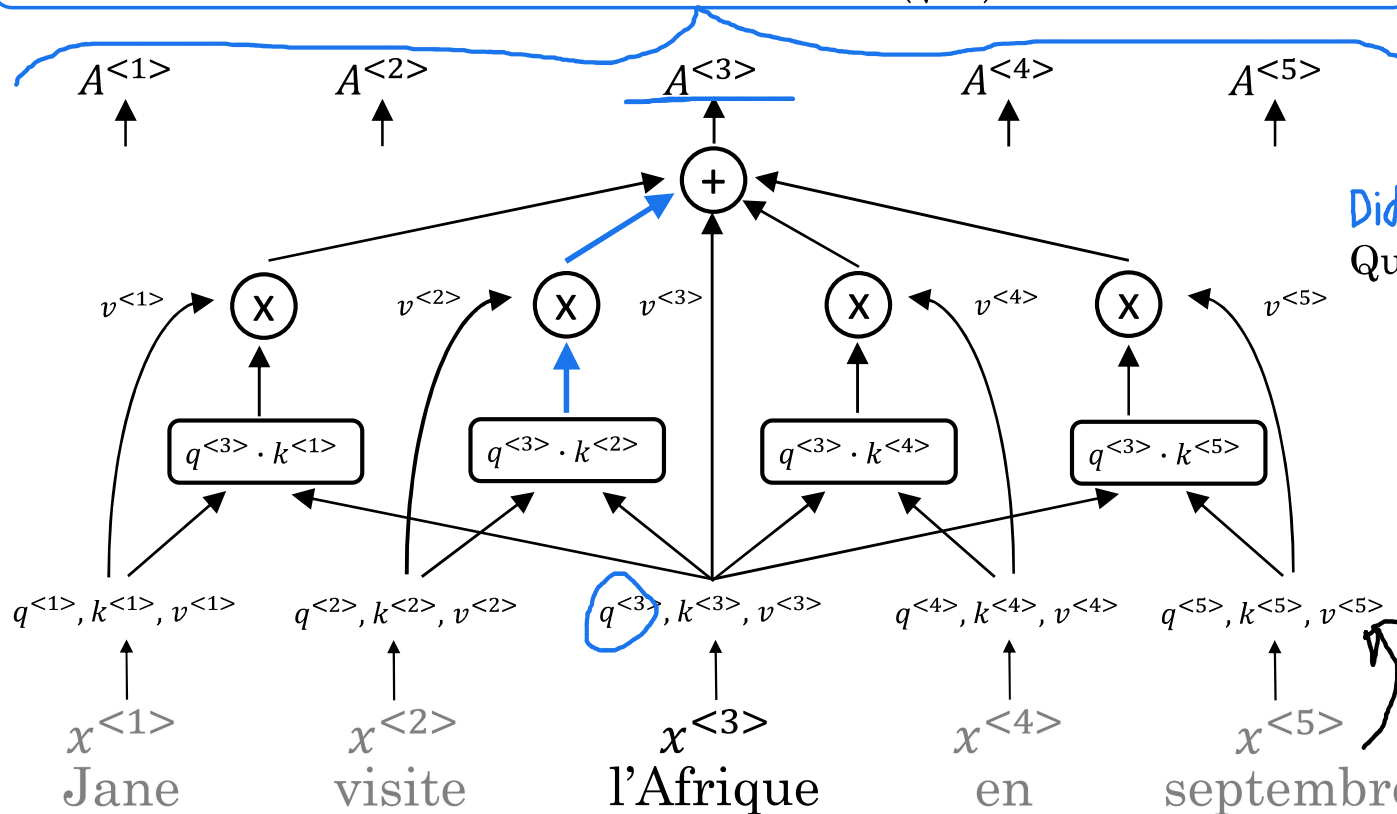
$$A(q, K, V) = \sum_i \frac{\exp(q \cdot k^{<i>})}{\sum_j \exp(q \cdot k^{<j>})} v^{<i>}$$

Self-Attention

$$A(q, K, V) = \sum_i \frac{\exp(e^{q \cdot k^{<i>}})}{\sum_j \exp(e^{q \cdot k^{<j>}})} v^{<i>}$$

Handwritten notes: "softmax" above the fraction, and blue circles around $q \cdot k^{<i>}$ and $q \cdot k^{<j>}$.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Did what?

Query (Q)

Key (K)

Value (V)

$q^{<1>}$

$k^{<1>}$

$v^{<1>}$

$q^{<2>}$

$k^{<2>}$

$v^{<2>}$

$q^{<3>}$

$k^{<3>}$

$v^{<3>}$

$q^{<4>}$

$k^{<4>}$

$v^{<4>}$

$q^{<5>}$

$k^{<5>}$

$v^{<5>}$

What's happening there?

person
action
Jane
visit

W^Q, W^K, W^V

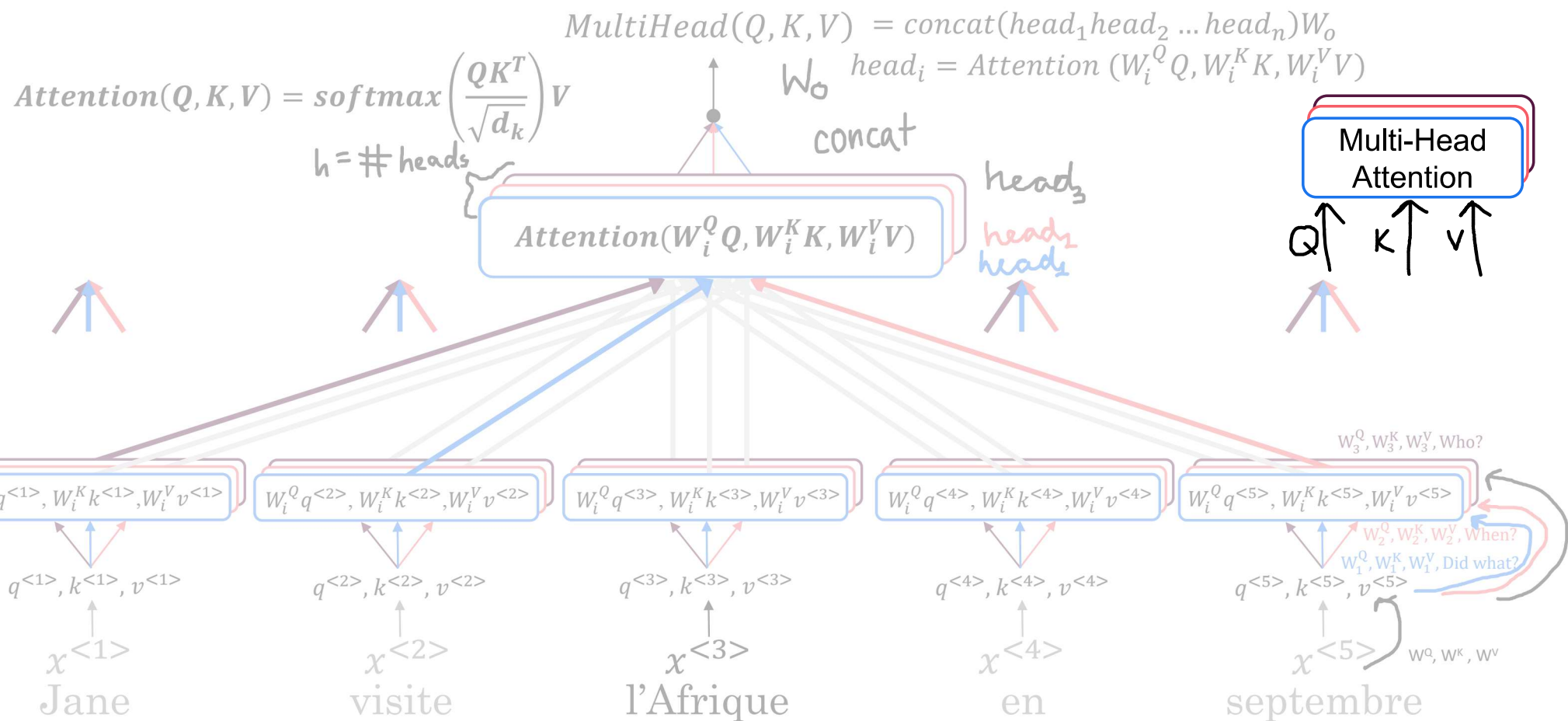


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Sequence to sequence models

Multi-Head Attention

Multi-Head Attention



[Vaswani et al. 2017, Attention Is All You Need]

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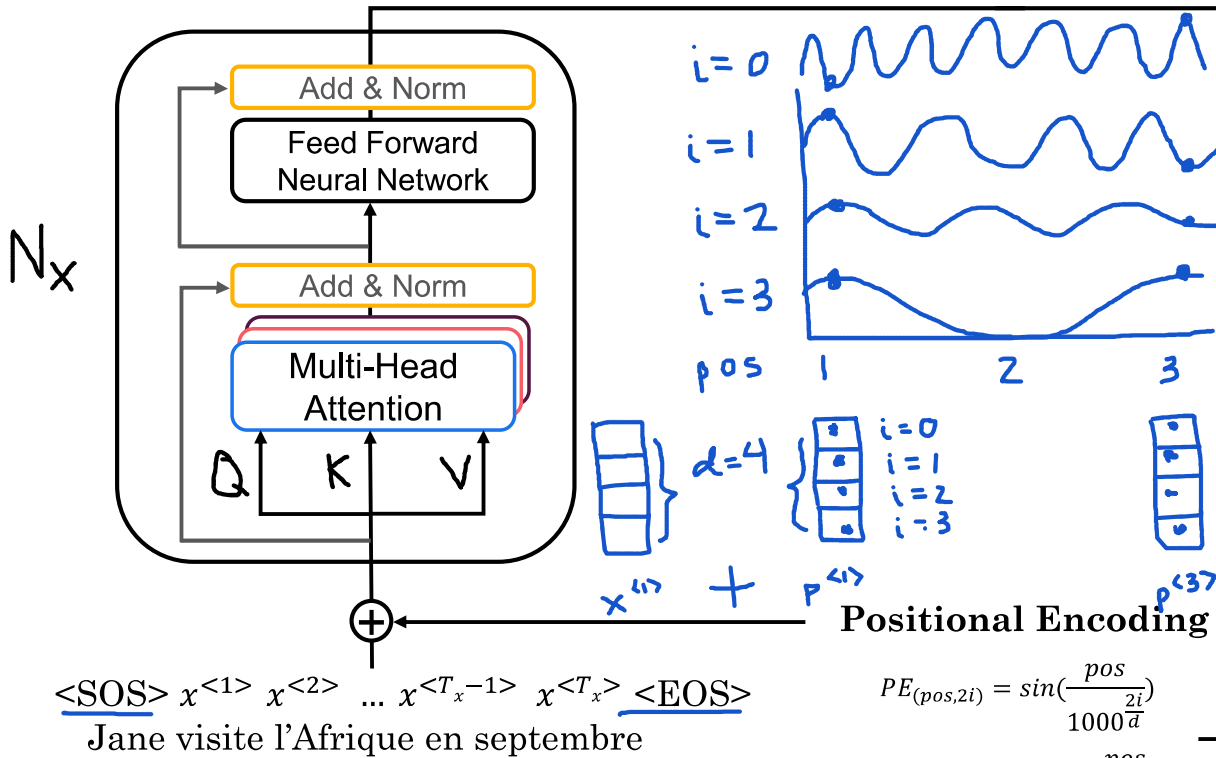
Sequence to sequence models

Transformers

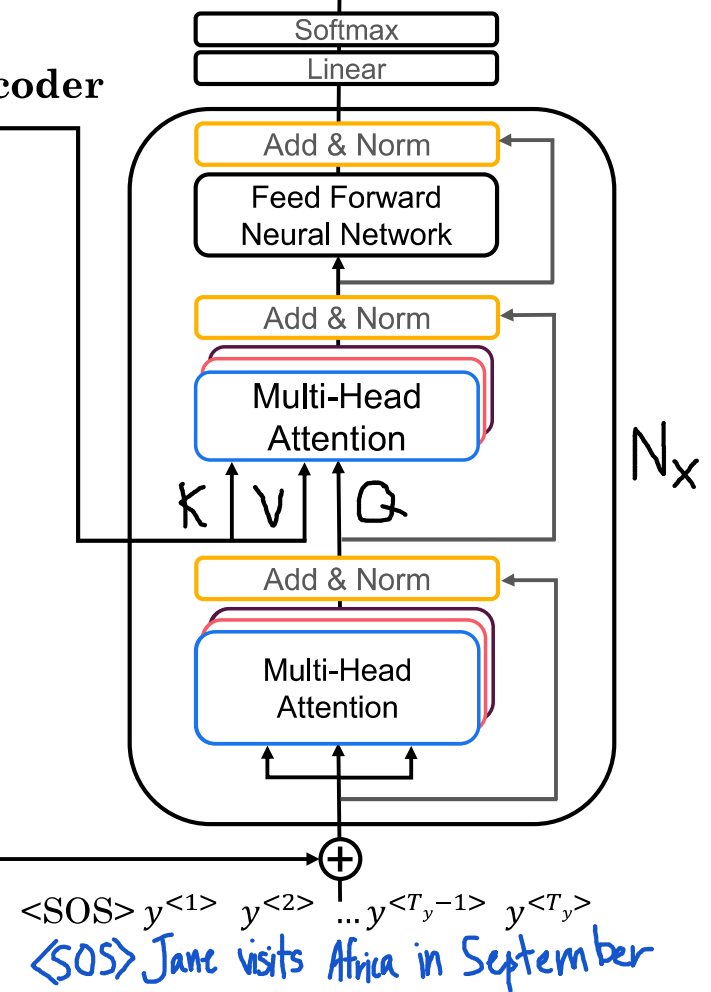
Transformer Details

<SOS> Jane visits Africa in September <EOS>

Encoder



Decoder



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